Worksheet No. – 01

Aim/Overview of the practical: The aim of the experiment is to create a worksheet with different data and apply the logical formulas on the data like IF Statement, Nested IF, AND, OR, NOT, IFERROR, SUMIF, AVERAGEIF, COUNTIF, and AVERAGEIF. These formulas allow you to add logic to your spreadsheet.

Objective: The objective of this experiment is to understand and apply various logical functions in Microsoft Excel for efficient data analysis and decision-making. It aims to create a worksheet with different types of data and use formulas such as IF, Nested IF, AND, OR, NOT, IFERROR, SUMIF, COUNTIF, and AVERAGEIF to perform conditional calculations, evaluate multiple criteria, handle errors, and summarize data accurately.

Procedure/Algorithm/Code

Procedure:

1. Open Microsoft Excel and create a new worksheet.
2. Enter the student data (Attendance and marks of 5 subjects).
3. Format the worksheet with headings and borders.
4. Apply the required formulas (IF, Nested IF, AND, OR, NOT, IFERROR, SUMIF, AVERAGEIF, and COUNTIF).
5. Verify the calculated results.
6. Save the worksheet.

IF FORMULA:

Definition:

The IF function checks whether a condition is TRUE or FALSE and returns one value if TRUE and another value if FALSE.

Condition:

Marks more than 35.

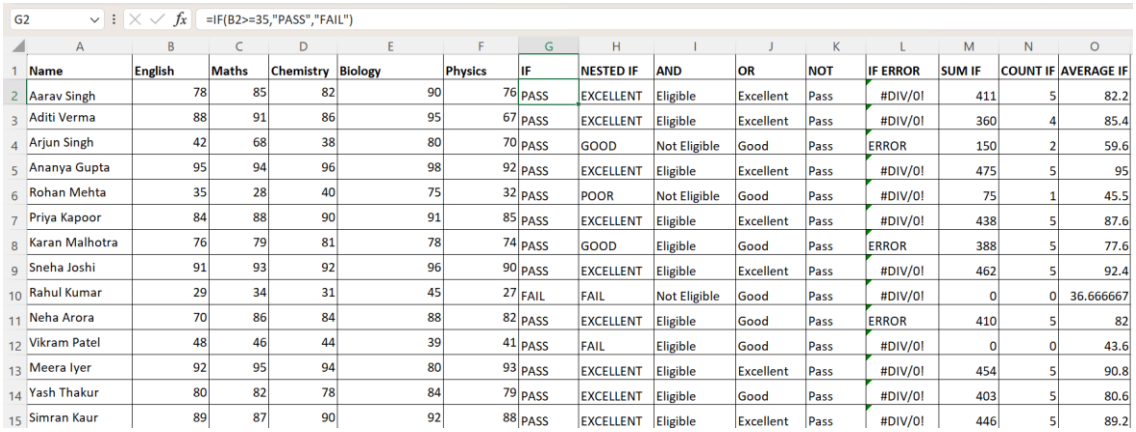
Formula:

=IF(B2>=35,"PASS","FAIL")

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25MCA-6 (B)

OUTPUT



=IF(B2>=35,"PASS","FAIL")

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Name	English	Maths	Chemistry	Biology	Physics	IF	NESTED IF	AND	OR	NOT	IF ERROR	SUM IF	COUNT IF	AVERAGE IF
2	Aarav Singh	78	85	82	90	76	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	411	5	82.2
3	Aditi Verma	88	91	86	95	67	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	360	4	85.4
4	Arjun Singh	42	68	38	80	70	PASS	GOOD	Not Eligible	Good	Pass	ERROR	150	2	59.6
5	Ananya Gupta	95	94	96	98	92	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	475	5	95
6	Rohan Mehta	35	28	40	75	32	PASS	POOR	Not Eligible	Good	Pass	#DIV/0!	75	1	45.5
7	Priya Kapoor	84	88	90	91	85	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	438	5	87.6
8	Karan Malhotra	76	79	81	78	74	PASS	GOOD	Eligible	Good	Pass	ERROR	388	5	77.6
9	Sneha Joshi	91	93	92	96	90	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	462	5	92.4
10	Rahul Kumar	29	34	31	45	27	FAIL	FAIL	Not Eligible	Good	Pass	#DIV/0!	0	0	36.666667
11	Neha Arora	70	86	84	88	82	PASS	EXCELLENT	Eligible	Good	Pass	ERROR	410	5	82
12	Vikram Patel	48	46	44	39	41	PASS	FAIL	Eligible	Good	Pass	#DIV/0!	0	0	43.6
13	Meera Iyer	92	95	94	80	93	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	454	5	90.8
14	Yash Thakur	80	82	78	84	79	PASS	EXCELLENT	Eligible	Good	Pass	#DIV/0!	403	5	80.6
15	Simran Kaur	89	87	90	92	88	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	446	5	89.2

NESTEDIF FORMULA:

Definition:

A Nested IF function contains multiple IF statements to evaluate several conditions in sequence.

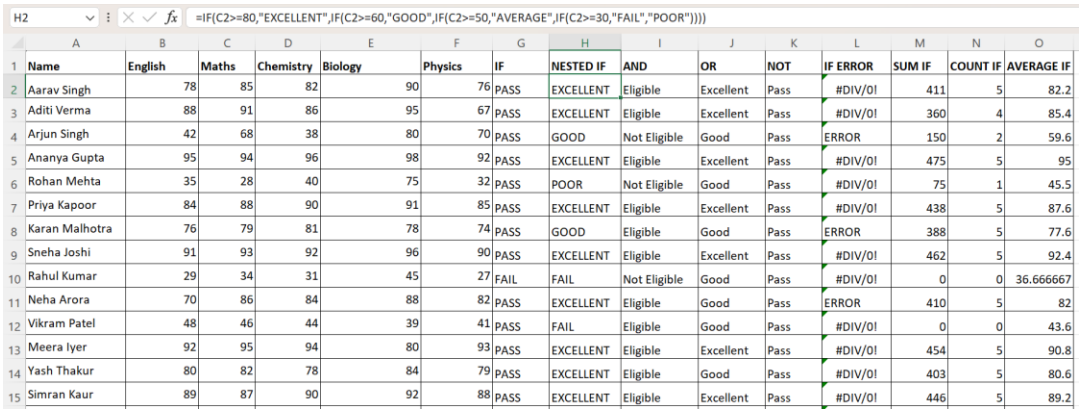
Condition: Condition based on marks:

- 80 or above → Excellent
- 79-60 → Good
- 59-50 → Average
- Below 30 → Fail

Formula:

=IF(C2>=80,"EXCELLENT",IF(C2>=60,"GOOD",IF(C2>=50,"AVERAGE",IF(C2>=30,"FAIL","POOR"))))

OUTPUT



=IF(C2>=80,"EXCELLENT",IF(C2>=60,"GOOD",IF(C2>=50,"AVERAGE",IF(C2>=30,"FAIL","POOR"))))

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Name	English	Maths	Chemistry	Biology	Physics	IF	NESTED IF	AND	OR	NOT	IF ERROR	SUM IF	COUNT IF	AVERAGE IF
2	Aarav Singh	78	85	82	90	76	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	411	5	82.2
3	Aditi Verma	88	91	86	95	67	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	360	4	85.4
4	Arjun Singh	42	68	38	80	70	PASS	GOOD	Not Eligible	Good	Pass	ERROR	150	2	59.6
5	Ananya Gupta	95	94	96	98	92	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	475	5	95
6	Rohan Mehta	35	28	40	75	32	PASS	POOR	Not Eligible	Good	Pass	#DIV/0!	75	1	45.5
7	Priya Kapoor	84	88	90	91	85	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	438	5	87.6
8	Karan Malhotra	76	79	81	78	74	PASS	GOOD	Eligible	Good	Pass	ERROR	388	5	77.6
9	Sneha Joshi	91	93	92	96	90	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	462	5	92.4
10	Rahul Kumar	29	34	31	45	27	FAIL	FAIL	Not Eligible	Good	Pass	#DIV/0!	0	0	36.666667
11	Neha Arora	70	86	84	88	82	PASS	EXCELLENT	Eligible	Good	Pass	ERROR	410	5	82
12	Vikram Patel	48	46	44	39	41	PASS	FAIL	Eligible	Good	Pass	#DIV/0!	0	0	43.6
13	Meera Iyer	92	95	94	80	93	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	454	5	90.8
14	Yash Thakur	80	82	78	84	79	PASS	EXCELLENT	Eligible	Good	Pass	#DIV/0!	403	5	80.6
15	Simran Kaur	89	87	90	92	88	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	446	5	89.2

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AND FORMULA:

Definition:

The AND function returns TRUE only if all specified conditions are TRUE.

Condition: Marks in Maths and Chemistry more than 40.

Formula: =IF(AND(C2>=40,D2>=40),"Eligible","Not Eligible")

OUTPUT

I2 =IF(AND(C2>=40,D2>=40),"Eligible","Not Eligible")

Name	English	Maths	Chemistry	Biology	Physics	IF	NESTED IF	AND	OR	NOT	IF ERROR	SUM IF	COUNT IF	AVERAGE IF
Aarav Singh	78	85	82	90	76	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	411	5	82.2
Aditi Verma	88	91	86	95	67	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	360	4	85.4
Arjun Singh	42	68	38	80	70	PASS	GOOD	Not Eligible	Good	Pass	ERROR	150	2	59.6
Ananya Gupta	95	94	96	98	92	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	475	5	95
Rohan Mehta	35	28	40	75	32	PASS	POOR	Not Eligible	Good	Pass	#DIV/0!	75	1	45.5
Priya Kapoor	84	88	90	91	85	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	438	5	87.6
Karan Malhotra	76	79	81	78	74	PASS	GOOD	Eligible	Good	Pass	ERROR	388	5	77.6
Sneha Joshi	91	93	92	96	90	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	462	5	92.4
Rahul Kumar	29	34	31	45	27	FAIL	FAIL	Not Eligible	Good	Pass	#DIV/0!	0	0	36.666667
Neha Arora	70	86	84	88	82	PASS	EXCELLENT	Eligible	Good	Pass	ERROR	410	5	82
Vikram Patel	48	46	44	39	41	PASS	FAIL	Eligible	Good	Pass	#DIV/0!	0	0	43.6
Meera Iyer	92	95	94	80	93	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	454	5	90.8
Yash Thakur	80	82	78	84	79	PASS	EXCELLENT	Eligible	Good	Pass	#DIV/0!	403	5	80.6
Simran Kaur	89	87	90	92	88	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	446	5	89.2

OR FORMULA:

Definition: The OR function returns TRUE if at least one of the specified conditions is TRUE.

Condition: Marks in Biology or Physics should more than 90.

Formula: =IF(OR(E2>=90,F2>=90),"Excellent","Good")

OUTPUT

J2 =IF(OR(E2>=90,F2>=90),"Excellent","Good")

Name	English	Maths	Chemistry	Biology	Physics	IF	NESTED IF	AND	OR	NOT	IF ERROR	SUM IF	COUNT IF	AVERAGE IF
Aarav Singh	78	85	82	90	76	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	411	5	82.2
Aditi Verma	88	91	86	95	67	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	360	4	85.4
Arjun Singh	42	68	38	80	70	PASS	GOOD	Not Eligible	Good	Pass	ERROR	150	2	59.6
Ananya Gupta	95	94	96	98	92	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	475	5	95
Rohan Mehta	35	28	40	75	32	PASS	POOR	Not Eligible	Good	Pass	#DIV/0!	75	1	45.5
Priya Kapoor	84	88	90	91	85	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	438	5	87.6
Karan Malhotra	76	79	81	78	74	PASS	GOOD	Eligible	Good	Pass	ERROR	388	5	77.6
Sneha Joshi	91	93	92	96	90	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	462	5	92.4
Rahul Kumar	29	34	31	45	27	FAIL	FAIL	Not Eligible	Good	Pass	#DIV/0!	0	0	36.666667
Neha Arora	70	86	84	88	82	PASS	EXCELLENT	Eligible	Good	Pass	ERROR	410	5	82
Vikram Patel	48	46	44	39	41	PASS	FAIL	Eligible	Good	Pass	#DIV/0!	0	0	43.6
Meera Iyer	92	95	94	80	93	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	454	5	90.8
Yash Thakur	80	82	78	84	79	PASS	EXCELLENT	Eligible	Good	Pass	#DIV/0!	403	5	80.6
Simran Kaur	89	87	90	92	88	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	446	5	89.2

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ERRORIF FORMULA:

Definition: The IFERROR function returns a specified value if a formula results in an error; otherwise, it returns the formula result.

Condition: If D2 ÷ 0 is TRUE (i.e., no error occurs), display ERROR otherwise, display NO ERROR.

Formula: =IF(D2/0,"ERROR","NO ERROR")

OUTPUT

L2 =IF(D2/0,"ERROR","NO ERROR")

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Name	English	Maths	Chemistry	Biology	Physics	IF	NESTED IF	AND	OR	NOT	IF ERROR	SUM IF	COUNT IF	AVERAGE IF
2	Aarav Singh	78	85	82	90	76	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	411	5	82.2
3	Aditi Verma	88	91	86	95	67	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	360	4	85.4
4	Arjun Singh	42	68	38	80	70	PASS	GOOD	Not Eligible	Good	Pass	ERROR	150	2	59.6
5	Ananya Gupta	95	94	96	98	92	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	475	5	95
6	Rohan Mehta	35	28	40	75	32	PASS	POOR	Not Eligible	Good	Pass	#DIV/0!	75	1	45.5
7	Priya Kapoor	84	88	90	91	85	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	438	5	87.6
8	Karan Malhotra	76	79	81	78	74	PASS	GOOD	Eligible	Good	Pass	ERROR	388	5	77.6
9	Sneha Joshi	91	93	92	96	90	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	462	5	92.4
10	Rahul Kumar	29	34	31	45	27	FAIL	FAIL	Not Eligible	Good	Pass	#DIV/0!	0	0	36.666667
11	Neha Arora	70	86	84	88	82	PASS	EXCELLENT	Eligible	Good	Pass	ERROR	410	5	82
12	Vikram Patel	48	46	44	39	41	PASS	FAIL	Eligible	Good	Pass	#DIV/0!	0	0	43.6
13	Meera Iyer	92	95	94	80	93	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	454	5	90.8
14	Yash Thakur	80	82	78	84	79	PASS	EXCELLENT	Eligible	Good	Pass	#DIV/0!	403	5	80.6
15	Simran Kaur	89	87	90	92	88	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	446	5	89.2

SUMIF FORMULA:

Definition: The SUMIF function adds only those values that satisfy a specified condition.

Condition: Add the marks of all subjects that are greater than or equal to 70.

Formula: =SUMIF(B2:F2,">=70",B2:F2)

OUTPUT

M2 =SUMIF(B2:F2,">=70",B2:F2)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Name	English	Maths	Chemistry	Biology	Physics	IF	NESTED IF	AND	OR	NOT	IF ERROR	SUM IF	COUNT IF	AVERAGE IF
2	Aarav Singh	78	85	82	90	76	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	411	5	82.2
3	Aditi Verma	88	91	86	95	67	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	360	4	85.4
4	Arjun Singh	42	68	38	80	70	PASS	GOOD	Not Eligible	Good	Pass	ERROR	150	2	59.6
5	Ananya Gupta	95	94	96	98	92	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	475	5	95
6	Rohan Mehta	35	28	40	75	32	PASS	POOR	Not Eligible	Good	Pass	#DIV/0!	75	1	45.5
7	Priya Kapoor	84	88	90	91	85	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	438	5	87.6
8	Karan Malhotra	76	79	81	78	74	PASS	GOOD	Eligible	Good	Pass	ERROR	388	5	77.6
9	Sneha Joshi	91	93	92	96	90	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	462	5	92.4
10	Rahul Kumar	29	34	31	45	27	FAIL	FAIL	Not Eligible	Good	Pass	#DIV/0!	0	0	36.666667
11	Neha Arora	70	86	84	88	82	PASS	EXCELLENT	Eligible	Good	Pass	ERROR	410	5	82
12	Vikram Patel	48	46	44	39	41	PASS	FAIL	Eligible	Good	Pass	#DIV/0!	0	0	43.6
13	Meera Iyer	92	95	94	80	93	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	454	5	90.8
14	Yash Thakur	80	82	78	84	79	PASS	EXCELLENT	Eligible	Good	Pass	#DIV/0!	403	5	80.6
15	Simran Kaur	89	87	90	92	88	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	446	5	89.2

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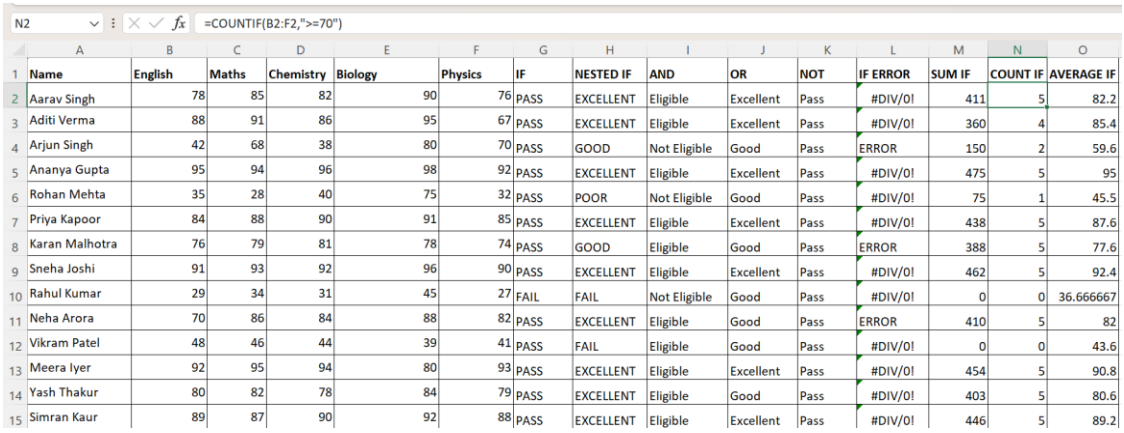
COUNTIF FORMULA:

Definition: The COUNTIF function counts the number of cells that meet a specified condition.

Condition: Count the number of subjects in which the marks are greater than or equal to 70.

Formula: =COUNTIF(B2:F2,">=70")

OUTPUT



N2 =COUNTIF(B2:F2,">=70")

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Name	English	Maths	Chemistry	Biology	Physics	IF	NESTED IF	AND	OR	NOT	IF ERROR	SUM IF	COUNT IF	AVERAGE IF
2	Aarav Singh	78	85	82	90	76	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	411	5	82.2
3	Aditi Verma	88	91	86	95	67	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	360	4	85.4
4	Arjun Singh	42	68	38	80	70	PASS	GOOD	Not Eligible	Good	Pass	ERROR	150	2	59.6
5	Ananya Gupta	95	94	96	98	92	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	475	5	95
6	Rohan Mehta	35	28	40	75	32	PASS	POOR	Not Eligible	Good	Pass	#DIV/0!	75	1	45.5
7	Priya Kapoor	84	88	90	91	85	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	438	5	87.6
8	Karan Malhotra	76	79	81	78	74	PASS	GOOD	Eligible	Good	Pass	ERROR	388	5	77.6
9	Sneha Joshi	91	93	92	96	90	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	462	5	92.4
10	Rahul Kumar	29	34	31	45	27	FAIL	FAIL	Not Eligible	Good	Pass	#DIV/0!	0	0	36.666667
11	Neha Arora	70	86	84	88	82	PASS	EXCELLENT	Eligible	Good	Pass	ERROR	410	5	82
12	Vikram Patel	48	46	44	39	41	PASS	FAIL	Eligible	Good	Pass	#DIV/0!	0	0	43.6
13	Meera Iyer	92	95	94	80	93	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	454	5	90.8
14	Yash Thakur	80	82	78	84	79	PASS	EXCELLENT	Eligible	Good	Pass	#DIV/0!	403	5	80.6
15	Simran Kaur	89	87	90	92	88	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	446	5	89.2

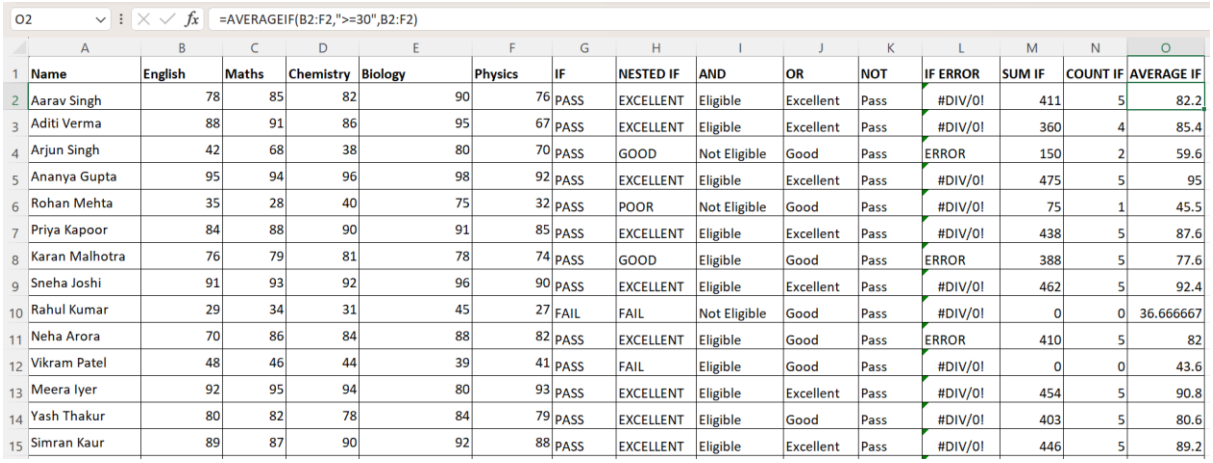
AVERAGEIF FORMULA:

Definition: The AVERAGEIF function calculates the average of values that meet a specified condition.

Condition: Calculate the average of the subject marks that are greater than 30.

Formula: =AVERAGEIF(B2:F2,">=30",B2:F2)

OUTPUT



O2 =AVERAGEIF(B2:F2,">=30",B2:F2)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Name	English	Maths	Chemistry	Biology	Physics	IF	NESTED IF	AND	OR	NOT	IF ERROR	SUM IF	COUNT IF	AVERAGE IF
2	Aarav Singh	78	85	82	90	76	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	411	5	82.2
3	Aditi Verma	88	91	86	95	67	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	360	4	85.4
4	Arjun Singh	42	68	38	80	70	PASS	GOOD	Not Eligible	Good	Pass	ERROR	150	2	59.6
5	Ananya Gupta	95	94	96	98	92	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	475	5	95
6	Rohan Mehta	35	28	40	75	32	PASS	POOR	Not Eligible	Good	Pass	#DIV/0!	75	1	45.5
7	Priya Kapoor	84	88	90	91	85	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	438	5	87.6
8	Karan Malhotra	76	79	81	78	74	PASS	GOOD	Eligible	Good	Pass	ERROR	388	5	77.6
9	Sneha Joshi	91	93	92	96	90	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	462	5	92.4
10	Rahul Kumar	29	34	31	45	27	FAIL	FAIL	Not Eligible	Good	Pass	#DIV/0!	0	0	36.666667
11	Neha Arora	70	86	84	88	82	PASS	EXCELLENT	Eligible	Good	Pass	ERROR	410	5	82
12	Vikram Patel	48	46	44	39	41	PASS	FAIL	Eligible	Good	Pass	#DIV/0!	0	0	43.6
13	Meera Iyer	92	95	94	80	93	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	454	5	90.8
14	Yash Thakur	80	82	78	84	79	PASS	EXCELLENT	Eligible	Good	Pass	#DIV/0!	403	5	80.6
15	Simran Kaur	89	87	90	92	88	PASS	EXCELLENT	Eligible	Excellent	Pass	#DIV/0!	446	5	89.2

Sukhleen Kaur (25MCA20031)
25MCA-6 (B)

Learning outcomes (What I have learnt):

1. Used logical formulas such as IF and Nested IF to perform conditional decision-making.
2. Used AND, OR, and NOT functions to evaluate multiple logical conditions.
3. Used formula for errors for effectively using the IFERROR function.
4. Used conditional calculations using SUMIF, AVERAGEIF, and COUNTIF functions.

Evaluation Grid:

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Demonstration and Performance		12
2.	Worksheet		8
3.	Viva		10